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Title: Transforming Cybersecurity with Agentic AI to Combat Emerging Cyber Threats

IEEE Citation

N. Kshetri, "Transforming Cybersecurity with Agentic AI to Combat Emerging Cyber Threats," in *Telecommunications Policy*, vol. 49, no. 6, art. no. 102976, 2025, doi: 10.1016/j.telpol.2025.102976.

Summary:

This paper explores the revolutionary capabilities of agentic AI in reshaping cybersecurity strategies to counter increasingly sophisticated cyber threats. Agentic AI, characterized by its autonomous decision-making and adaptive learning, is positioned as a key tool for automating essential functions in Security Operations Centers (SOCs), including real-time threat detection, incident response, and predictive analysis. The author argues that traditional cybersecurity approaches are inadequate against evolving dangers like advanced persistent threats and AI-generated attacks, and agentic AI can bridge this gap by enabling proactive, intelligent defenses. Through a review of current literature and case examples, the study demonstrates how agentic AI integrates with existing systems to enhance efficiency, such as by automating triage of alerts and orchestrating responses, potentially reducing human error and response times.

However, the integration of agentic AI introduces new vulnerabilities, including risks of adversarial attacks where malicious actors manipulate AI models, over-reliance on automation leading to skill degradation among human operators, and challenges in ensuring AI transparency and accountability. The paper discusses ethical concerns, such as bias in AI decision-making and potential privacy infringements from extensive data processing. It emphasizes the need to reassess established cybersecurity frameworks, like NIST or ISO standards, to incorporate AI-specific guidelines for governance, auditing, and resilience. Examples from industries like telecommunications illustrate both successes, such as AI-driven anomaly detection in network traffic, and pitfalls, like system failures from poorly trained models.

In conclusion, while agentic AI holds immense promise for transforming cybersecurity into a more dynamic and resilient field, the author calls for balanced implementation through robust risk management, continuous monitoring, and interdisciplinary collaboration. Recommendations include developing hybrid human-AI models in SOC and updating policy frameworks to mitigate risks, ensuring that AI advancements do not inadvertently create new attack vectors. The paper advocates for further research into agentic AI's long-term impacts, positioning it as essential for future-proofing digital infrastructures against emerging threats.